

HIV Analysis

STA198 Team: Natalie, Ella, Jessica, Ali

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```
rm(list = ls())
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.2.0      v readr      2.1.6
```

```
## v forcats   1.0.1      v stringr   1.6.0
```

```
## v ggplot2    4.0.2      v tibble    3.3.1
```

```
## v lubridate  1.9.5      v tidyr     1.3.2
```

```
## v purrr      1.2.1
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(janitor)
```

```
##
```

```
## Attaching package: 'janitor'
```

```
##
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
##      chisq.test, fisher.test
```

```
library(scales)
```

```
##
```

```
## Attaching package: 'scales'
```

```
##
```

```
## The following object is masked from 'package:purrr':
```

```
##
```

```
##      discard
```

```
##
```

```
## The following object is masked from 'package:readr':
```

```
##
```

```
##      col_factor
```

```
deaths <- read_csv("Data/no_of_deaths_by_country_clean.csv")
```

```
## Rows: 510 Columns: 7
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## chr (3): Country, Count, WHO Region
```

```
## dbl (4): Year, Count_median, Count_min, Count_max
```

```
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
hiv <- read_csv("Data/no_of_people_living_with_hiv_by_country_clean.csv")
```

```
## Rows: 680 Columns: 7
## -- Column specification -----
## Delimiter: ","
## chr (3): Country, Count, WHO Region
## dbl (4): Year, Count_median, Count_min, Count_max
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
art <- read_csv("Data/art_coverage_by_country_clean.csv")
```

```
## Rows: 170 Columns: 11
## -- Column specification -----
## Delimiter: ","
## chr (5): Country, Reported number of people receiving ART, Estimated number ...
## dbl (6): Estimated number of people living with HIV_median, Estimated number...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
deaths_clean <- deaths |>
  clean_names() |>
  group_by(country) |>
  summarise(
    deaths = first(count_median),
    .groups = "drop"
  )
```

```
hiv_clean <- hiv |>
  clean_names() |>
  group_by(country) |>
  summarise(
    hiv_population = first(count_median),
    .groups = "drop"
  )
```

```
art_clean <- art |>
  clean_names() |>
  select(
    country,
    art_coverage = estimated_art_coverage_among_people_living_with_hiv_percent_median
  )
```

```
final_df <- deaths_clean |>
  inner_join(hiv_clean, by = "country") |>
  inner_join(art_clean, by = "country")
```

```
write_csv(final_df, "Data/final_df.csv")
```

```
glimpse(final_df)
```

```
## Rows: 170
## Columns: 4
```

```
## $ country      <chr> "Afghanistan", "Albania", "Algeria", "Angola", "Argenti~
## $ deaths       <dbl> 500, NA, 200, 14000, 1700, 200, 200, NA, NA, 200, NA, 5~
## $ hiv_population <dbl> 7200, NA, 16000, 330000, 140000, 3500, 28000, NA, NA, 6~
## $ art_coverage  <dbl> 13, NA, 81, 27, 61, 53, 83, NA, NA, 52, NA, 22, 50, 59,~
```

Description

I created a merged, final dataset to use for analysis with cleaned and standardized columns (contains 170 countries and 4 variables). Each row represents a single country. The dataset was constructed by merging three country-level datasets sourced from WHO-based HIV/AIDS indicators available on Kaggle. All variables are continuous numerical estimates, and represent median values reported in the original datasets. The variables included in the final dataset are: - HIV/AIDS deaths (deaths) - Number of people living with HIV (hiv_population) - ART coverage among people living with HIV (art_coverage)

```
colSums(is.na(final_df))
```

```
##      country      deaths hiv_population  art_coverage
##           0          37           32           34
```

```
colMeans(is.na(final_df))
```

```
##      country      deaths hiv_population  art_coverage
## 0.0000000 0.2176471 0.1882353 0.2000000
```

Description

The dataset contains missing values across all three main variables: HIV/AIDS deaths (37 missing, ~21.8%), number of people living with HIV (32 missing, ~18.8%), and ART coverage (34 missing, 20%); no missing data for countries. This is most likely due to incomplete reporting in certain countries within the original WHO-based dataset (real-world data is often incomplete). Missing data was not cleaned or removed, just be wary of it and decide what to do with it in the next stages (if anything).

```
final_df |>
  arrange(desc(deaths)) |>
  slice(1:10)
```

```
## # A tibble: 10 x 4
##   country      deaths hiv_population art_coverage
##   <chr>      <dbl>      <dbl>      <dbl>
## 1 South Africa 71000      7700000      62
## 2 Mozambique  54000      2200000      56
## 3 Nigeria     53000      1900000      53
## 4 Indonesia   38000       640000      17
## 5 Kenya     25000      1600000      68
## 6 United Republic of Tanzania 24000      1600000      71
## 7 Uganda      23000      1400000      72
## 8 Zimbabwe    22000      1300000      88
## 9 Cameroon    18000       540000      52
## 10 Thailand   18000       480000      75
```

Description

Shows top 10 countries with the most HIV/AIDS deaths.

```
final_df |>
  filter(!is.na(deaths)) |>
```

```
arrange(deaths) |>
slice(1:10)
```

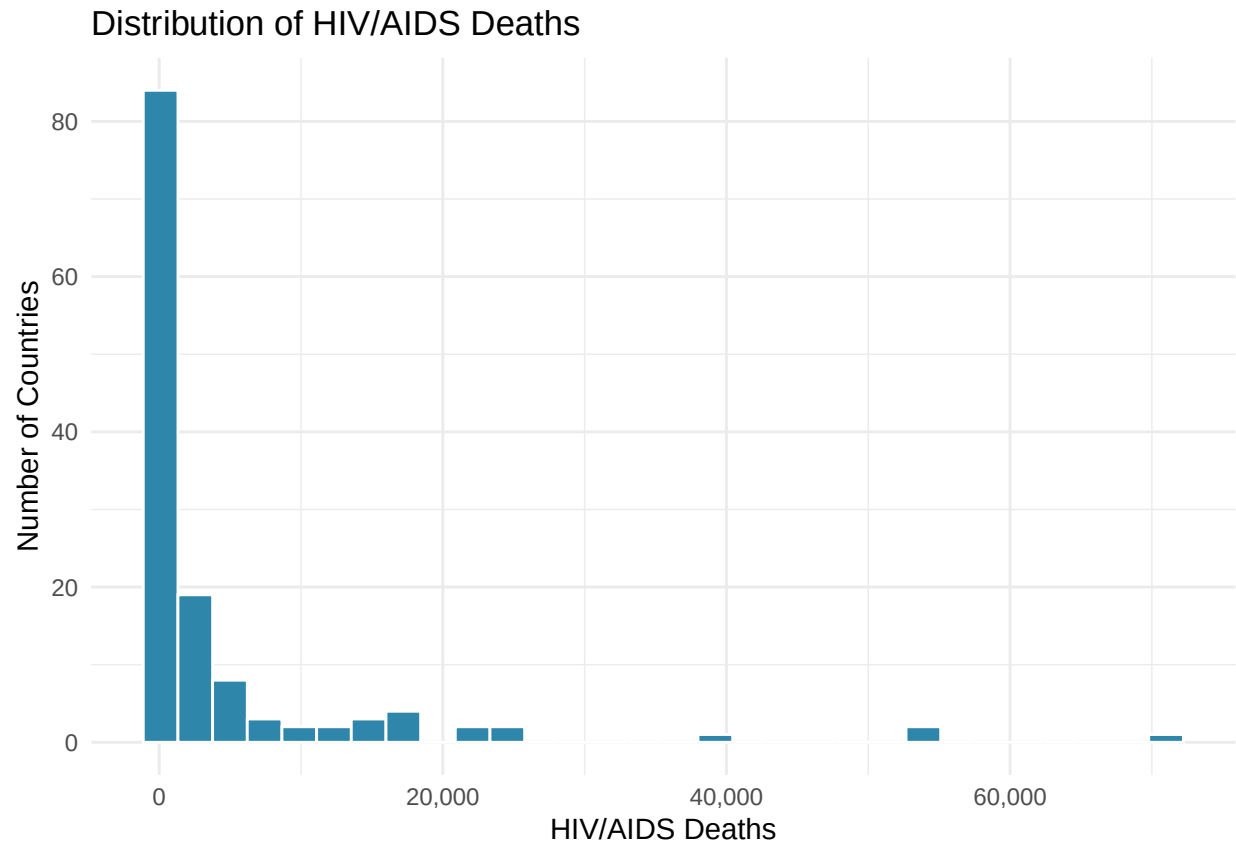
```
## # A tibble: 10 x 4
##   country      deaths hiv_population art_coverage
##   <chr>         <dbl>         <dbl>         <dbl>
## 1 Barbados         100         3000          50
## 2 Bhutan           100         1300          37
## 3 Bosnia and Herzegovina 100          500          67
## 4 Bulgaria         100         3500          41
## 5 Cabo Verde       100         2400          89
## 6 Comoros          100          200          79
## 7 Croatia          100         1600          75
## 8 Czechia          100         4400          60
## 9 Denmark          100         6200          89
## 10 Estonia         100         7400          59
```

Description

Shows top 10 countries with the least HIV/AIDS deaths.

```
ggplot(final_df, aes(x = deaths)) +
  geom_histogram(bins = 30, fill = "#2E86AB", color = "white") +
  scale_x_continuous(labels = label_comma()) +
  labs(
    title = "Distribution of HIV/AIDS Deaths",
    x = "HIV/AIDS Deaths",
    y = "Number of Countries"
  ) +
  theme_minimal()
```

```
## Warning: Removed 37 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

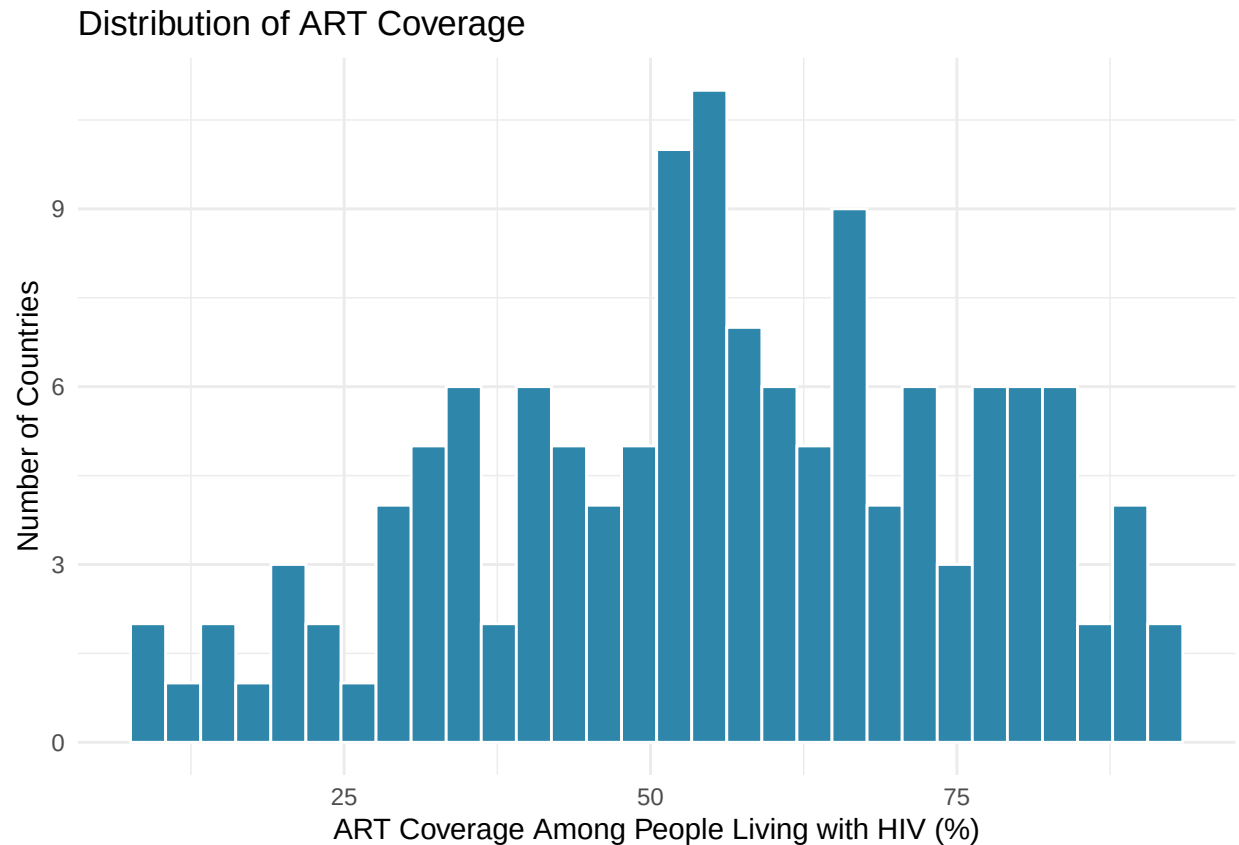


Description

Shows the distribution of deaths across countries.

```
ggplot(final_df, aes(x = art_coverage)) +
  geom_histogram(bins = 30, fill = "#2E86AB", color = "white") +
  labs(
    title = "Distribution of ART Coverage",
    x = "ART Coverage Among People Living with HIV (%)",
    y = "Number of Countries"
  ) +
  theme_minimal()
```

```
## Warning: Removed 34 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

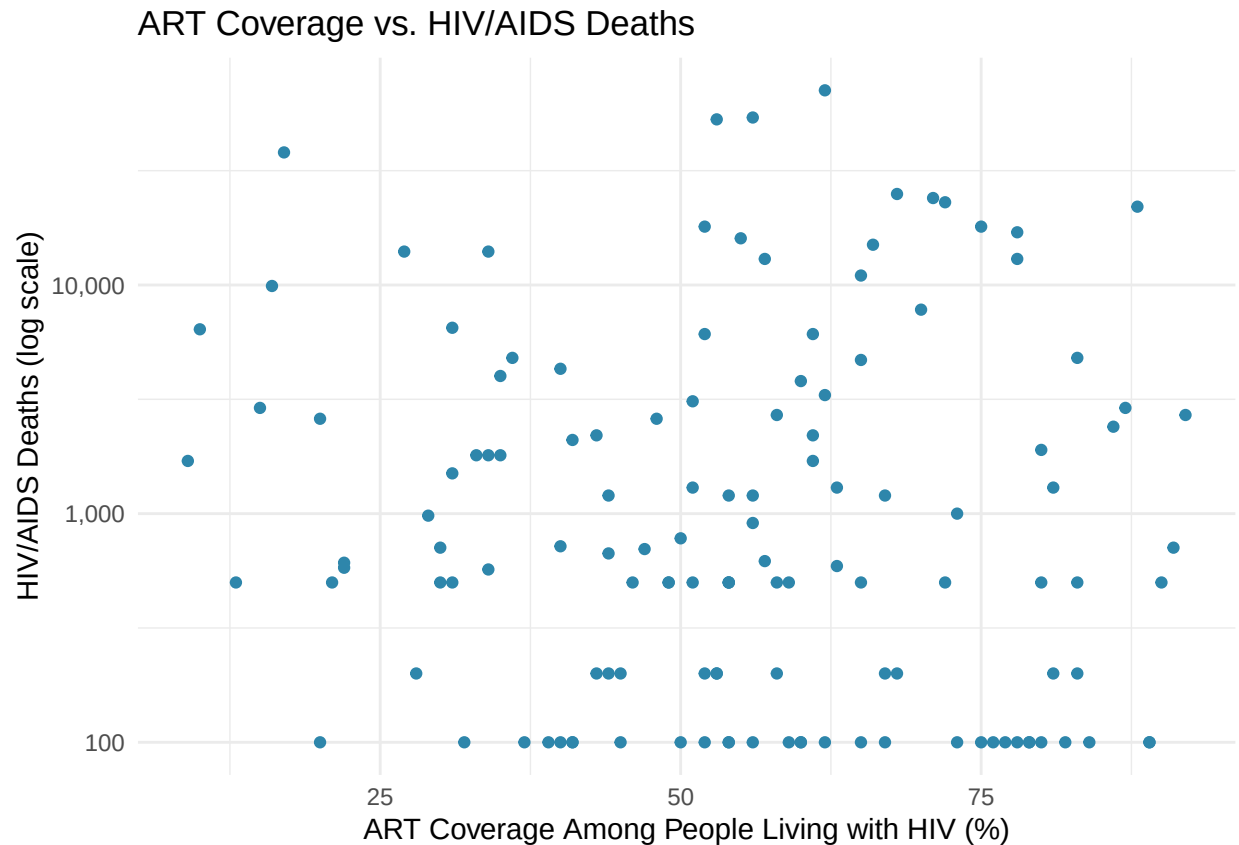


Description

Shows the distribution of treatment access (ART coverage %) across countries.

```
ggplot(final_df, aes(x = art_coverage, y = deaths)) +
  geom_point(color = "#2E86AB") +
  scale_y_log10(labels = label_comma()) +
  labs(
    title = "ART Coverage vs. HIV/AIDS Deaths",
    x = "ART Coverage Among People Living with HIV (%)",
    y = "HIV/AIDS Deaths (log scale)"
  ) +
  theme_minimal()
```

```
## Warning: Removed 39 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Description

Shows the relationship between ART coverage (%) and HIV/AIDS deaths across countries, where each point represents a country. Note that I used log scaling for the y-axis to account for scaling issues and compression of the graph.

Model Building

To examine the relationship between ART coverage and HIV/AIDS deaths, I conducted a correlation analysis and multiple linear regression models. I first tested the direct association between ART coverage and deaths, then included the number of people living with HIV as a control variable.

Correlational Test

```
cor(final_df$art_coverage, final_df$deaths, use = "complete.obs")
```

```
## [1] 0.02297943
```

Statistical Analysis:

The correlation between ART coverage and HIV/AIDS was very weak ($r = 0.023$), indicating little to no linear relationship when examined alone. However, this result may be misleading because it does not account for differences in the number of people living with HIV across countries, which can strongly influence death counts. This means that a controlled regression analysis should be conducted.

Simple regression model

```
model1 <- lm(deaths ~ art_coverage, data = final_df)
summary(model1)

##
## Call:
## lm(formula = deaths ~ art_coverage, data = final_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4934  -4400  -3800  -1538   66292
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3960.15    2697.69   1.468   0.145
## art_coverage    12.07     46.22   0.261   0.794
##
## Residual standard error: 10580 on 129 degrees of freedom
## (39 observations deleted due to missingness)
## Multiple R-squared:  0.0005281, Adjusted R-squared:  -0.00722
## F-statistic: 0.06815 on 1 and 129 DF,  p-value: 0.7945
```

Statistical Analysis:

A simple linear regression was conducted to examine the relationship between ART coverage and HIV/AIDS deaths. The results showed no significant relationship between ART coverage and deaths ($\beta = 12.07$, $p = 0.794$). Additionally, the model explained virtually none of the variation in deaths ($R^2 = 0.0005$), indicating that ART coverage alone is not a meaningful predictor of HIV/AIDS mortality. This suggests that other factors, such as the size of the HIV-positive population, may play a more important role.

Controlled regression model

```
model2 <- lm(deaths ~ art_coverage + hiv_population, data = final_df)
summary(model2)

##
## Call:
## lm(formula = deaths ~ art_coverage + hiv_population, data = final_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -23510.6  -1746.6  -1210.4   -276.6   28179.1
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   4.021e+03  1.347e+03   2.985   0.0034 **
## art_coverage  -4.055e+01  2.323e+01  -1.745   0.0833 .
## hiv_population  1.208e-02  6.121e-04  19.732 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5285 on 128 degrees of freedom
## (39 observations deleted due to missingness)
```

```
## Multiple R-squared:  0.7527, Adjusted R-squared:  0.7489
## F-statistic: 194.8 on 2 and 128 DF,  p-value: < 2.2e-16
```

Statistical Analysis:

A multiple linear regression model was used to examine the relationship between ART coverage and HIV/AIDS deaths while controlling for the number of people living with HIV. The results showed that ART coverage had a negative association with deaths ($\beta = -40.55$, $p = 0.083$), suggesting that higher ART coverage is associated with fewer HIV/AIDS deaths. Although this relationship was not statistically significant at the 0.05 level, it indicates a meaningful downward trend.

In contrast, the number of people living with HIV was a highly significant predictor of deaths ($\beta = 0.01208$, $p = 2 \times 10^{-16}$), indicating that countries with larger HIV-positive populations experience substantially more deaths.

The model overall demonstrated strong explanatory power ($R^2 = 0.753$), meaning that approximately 75% of the variation in HIV/AIDS deaths across countries is explained by these variables.

All in all, while ART coverage alone did not appear to influence deaths in the simple model, the controlled regression suggests that increased ART coverage is associated with reduced HIV/AIDS mortality when accounting for differences in HIV pop. size. This supports the hypothesis that expanding access to antiretroviral therapy may help reduce HIV-related deaths globally.

Log model

```
model3 <- lm(log(deaths) ~ art_coverage + log(hiv_population), data = final_df)
summary(model3)
```

```
##
## Call:
## lm(formula = log(deaths) ~ art_coverage + log(hiv_population),
##     data = final_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.61043 -0.38433  0.01353  0.34033  2.16313
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -0.662964   0.275172  -2.409   0.0174 *
## art_coverage   -0.014334   0.002538  -5.649 9.96e-08 ***
## log(hiv_population) 0.799767   0.023634  33.840 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5792 on 128 degrees of freedom
## (39 observations deleted due to missingness)
## Multiple R-squared:  0.9001, Adjusted R-squared:  0.8986
## F-statistic: 576.8 on 2 and 128 DF,  p-value: < 2.2e-16
```

Statistical Analysis:

A log-transformed multiple regression model was used to better account for skewness in the data and examine the relationship between ART coverage and HIV/AIDS deaths. The model takes the form:

$$\log(\text{deaths}) = \beta_0 + \beta_1(\text{art_coverage}) + \beta_2 \cdot \log(\text{hiv_population}) +$$

The results showed that ART coverage had a statistically significant negative association with log(HIV/AIDS deaths) ($\beta = -0.0143$, $p = 9.96e-08$), after controlling for log(HIV population size). This means that for each 1 percentage point increase in ART coverage, a country's expected HIV/AIDS deaths are multiplied by a factor of $e^{(-0.0143)} = 0.986$. This is about a 1.4% decrease in deaths per additional percentage point of ART coverage.

log(HIV population size) was also highly significant ($\beta = 0.800$, $p < 2e-16$). This confirms that population size remains the main contributor to death counts. This model explained 90.0% of the variation in log(HIV/AIDS deaths) ($R^2 = 0.900$), a substantial improvement over both the simple ($R^2 = 0.0005$) and controlled ($R^2 = 0.753$) untransformed models, suggesting the log transformation better captures the underlying relationships in the data.

Graphs and Results

```
library(tidyverse)
library(broom)
library(scales)
library(knitr)

final_df <- read_csv("Data/final_df.csv")

## Rows: 170 Columns: 4
## -- Column specification -----
## Delimiter: ","
## chr (1): country
## dbl (3): deaths, hiv_population, art_coverage
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

final_df_clean <- final_df |>
  filter(!is.na(art_coverage), !is.na(deaths),
         !is.na(hiv_population), deaths > 0, hiv_population > 0)

model3 <- lm(log(deaths) ~ art_coverage + log(hiv_population),
             data = final_df_clean)
```

Scatterplot of ART coverage vs. log(deaths) with regression line and confidence band

```
ggplot(final_df_clean, aes(x = art_coverage, y = log(deaths))) +
  geom_point(color = "#2E86AB") +
  geom_smooth(method = lm, se = TRUE, color = "#E84855") +
  labs(
    title = "ART Coverage vs. Log HIV/AIDS Deaths",
    x = "ART Coverage Among People Living with HIV (%)",
    y = "log(HIV/AIDS Deaths)"
  ) +
  theme_minimal()

## `geom_smooth()` using formula = 'y ~ x'
```

ART Coverage vs. Log HIV/AIDS Deaths

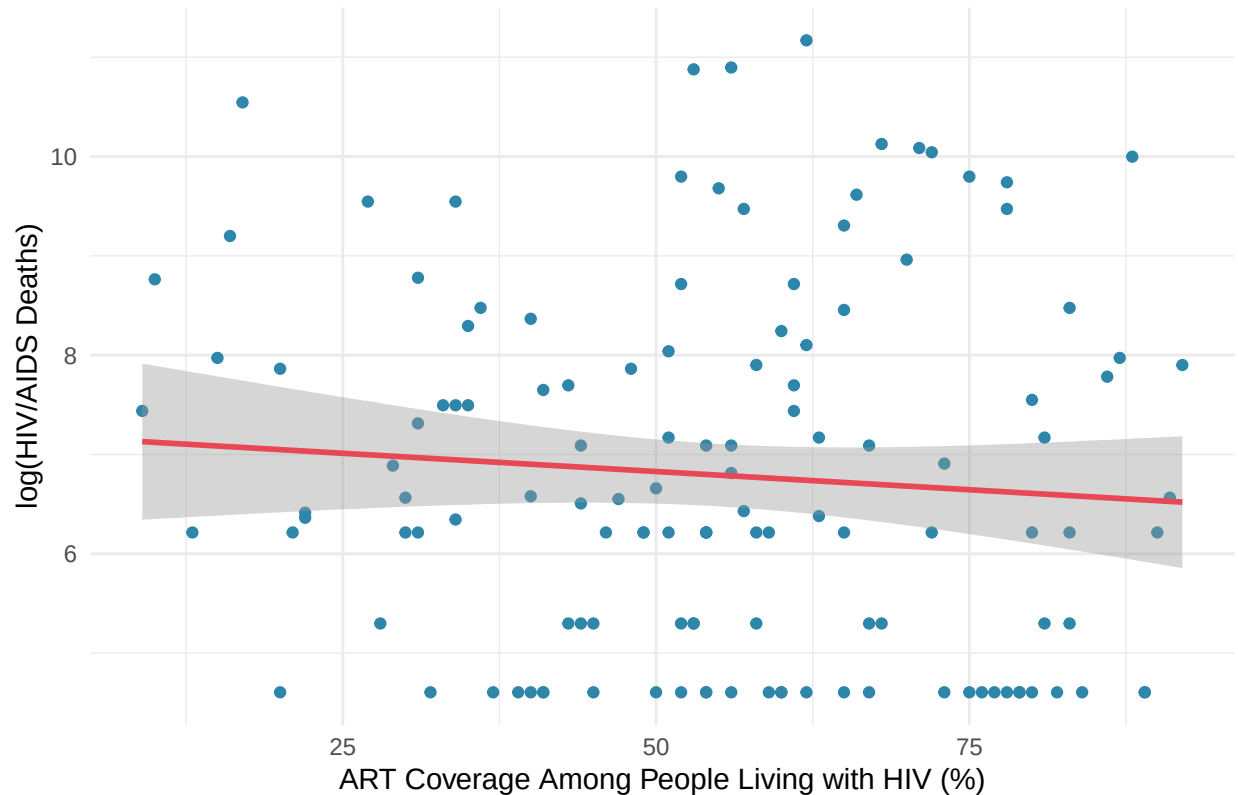


Figure 1 displays the relationship between ART coverage and log(HIV/AIDS deaths) across countries, with a fitted regression line from Model 3 and 95% confidence band. The negative slope visually shows that higher ART coverage is associated with lower HIV/AIDS mortality, while controlling for HIV population size.

Coefficient plot comparing the ART coverage estimate across all three models with 95% CI bars

```
m1_tidy <- tidy(model1, conf.int = TRUE) |> mutate(model = "Model 1\nSimple OLS")
m2_tidy <- tidy(model2, conf.int = TRUE) |> mutate(model = "Model 2\nControlled OLS")
m3_tidy <- tidy(model3, conf.int = TRUE) |> mutate(model = "Model 3\nLog-Log")

bind_rows(m1_tidy, m2_tidy, m3_tidy) |>
  filter(term == "art_coverage") |>
  ggplot(aes(x = estimate, y = model, color = model)) +
  geom_point(size = 3) +
  geom_errorbarh(aes(xmin = conf.low, xmax = conf.high), height = 0.2) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "gray40") +
  scale_color_manual(values = c("#2E86AB", "#F4A261", "#E84855")) +
  labs(
    title = "ART Coverage Coefficient Across Models",
    x = "Coefficient Estimate",
    y = NULL
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

```
## Warning: `geom_errorbarh()` was deprecated in ggplot2 4.0.0.
## i Please use the `orientation` argument of `geom_errorbar()` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

## `height` was translated to `width`.
```

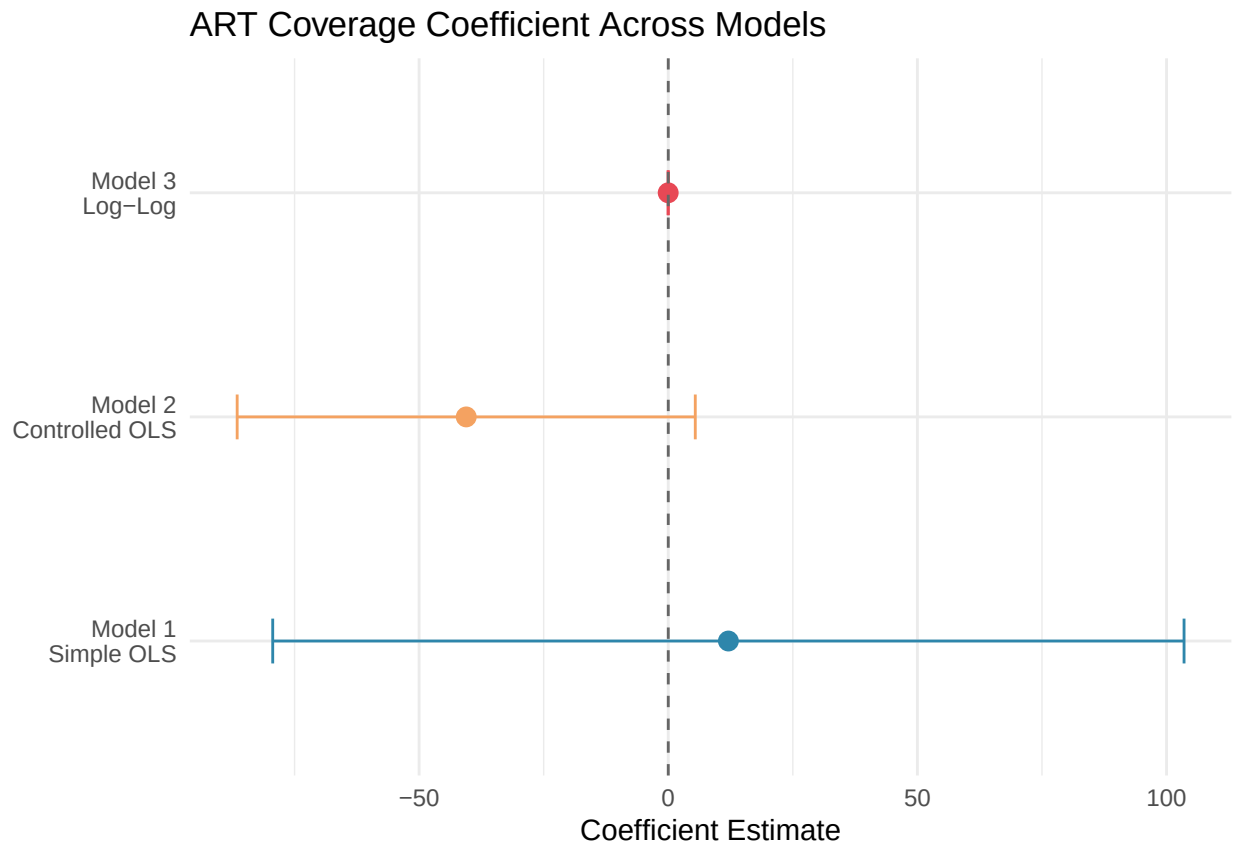


Figure 2 compares the ART coverage coefficient estimate across all three model specifications with 95% confidence intervals. In Models 1 and 2, the confidence interval crosses zero, indicating no statistically significant effect. Only in Model 3, after log-transforming both the outcome and HIV population size, does the estimate become significant, supporting the use of the log-log specification.

Residuals vs. Fitted for model3

```
augment(model3) |>
  ggplot(aes(x = .fitted, y = .resid)) +
  geom_point(color = "#2E86AB") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "#E84855") +
  labs(
    title = "Residuals vs. Fitted Values: Model 3",
    x = "Fitted Values",
    y = "Residuals"
  ) +
  theme_minimal()
```

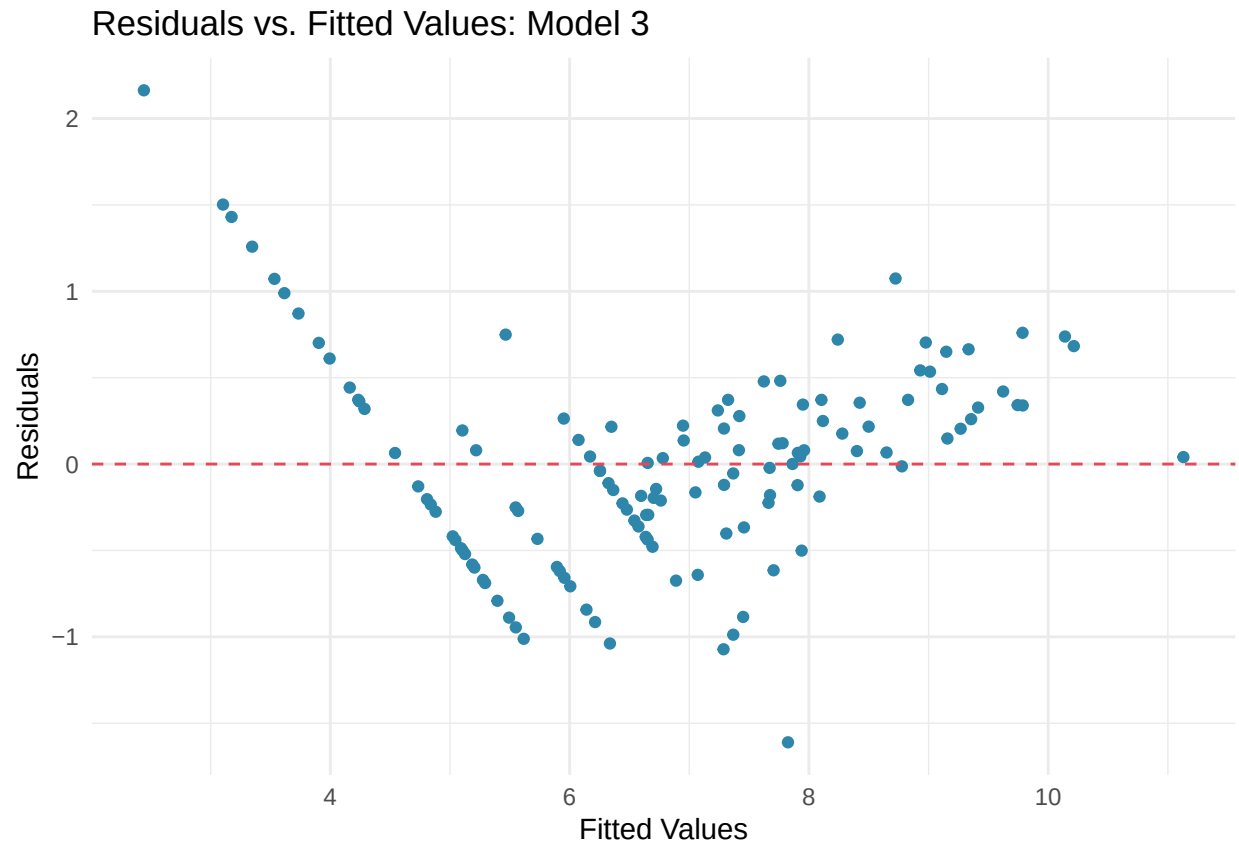


Figure 3 shows residuals scattered approximately randomly around zero with no strong systematic pattern, suggesting the linearity and constant variance assumptions of Model 3 are reasonably satisfied.

Histogram of Residuals for model3

```
augment(model3) |>
  ggplot(aes(x = .resid)) +
  geom_histogram(bins = 25, fill = "#2E86AB", color = "white") +
  labs(
    title = "Distribution of Residuals: Model 3",
    x = "Residuals",
    y = "Count"
  ) +
  theme_minimal()
```

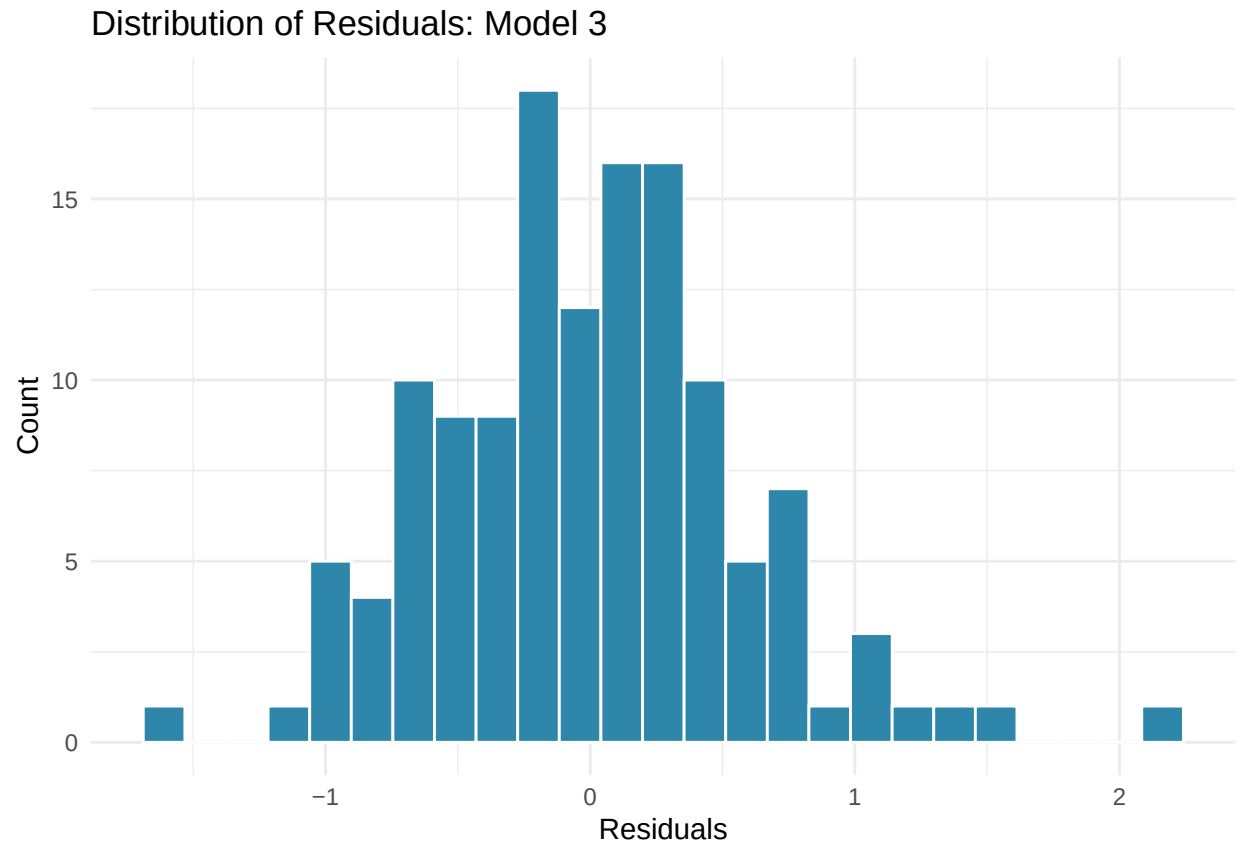


Figure 4 shows the distribution of residuals from Model 3. The approximately bell-shaped, symmetric distribution centered near zero suggests the normality assumption is reasonably met, supporting the validity of the model's inference.